PT VECTOR MANAGEMENT CONSULTING

PROPOSAL EXTENSION

ABL Operational Blueprinting Application Build Extension v4

Constraint-Governed Operational Scheduling Platform

Berau Coal Transshipment Operations

Prepared for	ABL Group
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Operating context	ABL - Berau Coal transshipment, port, barging, CTS and OGV loading operations
Document role	Application build extension to the Operational Blueprinting & Simulation Prototype Proposal
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Purpose: This extension defines how the operational blueprint will be translated into a client-facing application build. It frames the first release around governed scheduling and operating-window logic, while positioning scenario, telemetry, recovery and broader flow-management capabilities as phase-gated maturity blocks.

Client-facing application build extension

Contents

1. Executive Summary

This extension defines the application build direction for the ABL - Berau Coal operational blueprint. The proposed product is a **Constraint-Governed Operational Scheduling Platform** that converts shipment demand into a controlled, publishable logistics plan across cargo readiness, jetty/BLC loading, tug-barge movement, tide and bridge windows, CTS operations, OGV loading sequence, approval and plan publication.

The first release is deliberately focused. It will prove the governed scheduling spine and operating-window logic between Sprint 0 and Sprint 5. At this stage, the client receives a working scheduling application that replaces spreadsheet-only planning for the core ABL-Berau flow and creates an approved execution contract.

Vector will use its proprietary scheduling, scenario and dynamic-constraint scaffolding as the implementation base. The ABL-specific application will configure this base around Berau demand, ABL resources, operating windows, route eligibility, conflict handling, approvals and publishable schedule output.

Later capabilities such as exception recovery, scenario impact simulation, telemetry-assisted execution, confirmed field events and broader flow-management review will be introduced only after adoption, data readiness and governance behavior are validated.

2. Business Use Case And First-Release Objective

The business use case is:

Operating principle: Synchronize Berau Coal shipment commitments with ABL operational capacity so that cargo readiness, asset availability, tide/bridge windows, jetty/CTS capacity, OGV loading sequence and recovery decisions are planned through one governed scheduling platform.

2.1 Business problems addressed

Business problem	Application response
Spreadsheet planning is difficult to govern when demand, windows and assets change during the day	Structured demand intake, versioned plan generation, conflict checks, approval and published schedule output
Shipment demand is not always synchronized with real route and asset capacity	Demand, cargo sequence, resource eligibility, jetty/BLC capacity, CTS operations and movement windows are handled in one planning model
Tide, bridge, berth and operational windows can become the controlling constraint	Operating windows are treated as schedulable capacity, not as static notes
Cargo grade and loading sequence affect execution	Cargo requirements and loading sequence are captured as planning inputs
Active trips and occupied resources affect new scheduling decisions	Work in progress, occupied windows, unavailable assets and queues are plotted before new movements are placed
Planners need discretion for negotiated exceptions	Manual overrides remain available but are reason-coded, auditable and visible in plan lineage
Approval and publication require traceability	Published plans carry version, approval, audit and export lineage

2.2 First-release objective

Release 1 will deliver a governed scheduling MVP. The product objective is to provide:

- ? one structured intake for OGV demand and cargo sequence;

- ? one operating model for assets, routes, tide windows, bridge windows, jetty/BLC capacity and CTS capacity;
- ? one draft-plan generation flow with explicit feasibility and conflict output;
- ? one approval and publish path that creates an execution contract;
- ? one controlled export and audit trail for operational use.

Release 1 does not depend on live telemetry, autonomous recovery, commercial settlement logic or automatic optimization.

3. Client Flow Challenges

The ABL-Berau flow is difficult to schedule because the planning constraint does not stay fixed. At one point the limiting factor may be cargo readiness. Later the same shipment may be blocked by a tide window, a bridge crossing slot, a tug-barge availability issue, a CTS queue or an OGV laycan pressure.

The application is therefore designed around a moving-constraint operating flow.

Flow challenge	Planning implication
OGV demand changes or sequence changes	The schedule must be revalidated against cargo, route and downstream commitments
Cargo may be ready but an asset may not be available	Resource eligibility and availability must be checked before movement placement
A tug-barge movement may miss tide or bridge timing	The movement chain must be placed against continuous feasible windows
Jetty/BLC operations may delay barge loading	Upstream delays must be visible before downstream movements are committed
CTS or floating equipment may become the bottleneck	Discharge and transshipment capacity must be represented as a schedulable constraint
Planners may negotiate practical exceptions	The tool must preserve planner discretion while capturing reason and approval lineage
Live signals may be incomplete or stale	Field evidence must be separated from confirmed schedule authority

4. Scheduling-First Operating Philosophy

The scheduling philosophy is simple:

> Each shipment movement is treated as schedulable work that must pass through eligible route steps, available resources and valid operating windows.

For ABL, the route is both physical and operational:

```
Cargo readiness
-> Jetty / BLC loading
-> Tug-barge departure
-> Tide / river movement
-> Bridge crossing
-> CTS / anchorage queue
-> CTS discharge
-> OGV hatch / layer completion
-> Survey / handover closure
```

The first release will not overcomplicate this philosophy. It will focus on making the planning chain structured, visible, governable and publishable. Later maturity blocks can add richer scenario, recovery and live-execution behavior after the governed scheduling spine is adopted.

5. Vector Proprietary Scheduling And Scenario Scaffolding

Vector owns proprietary scaffolding for constraint-aware scheduling, scenario simulation and dynamic-capacity planning. This base will be used for the application build so that the client implementation can focus on ABL-specific operating rules, user workflows, data interfaces and adoption.

The operator does not need to see the internal engine mechanics. The visible behavior of the tool is:

? identify whether a movement is schedulable;

? surface the first meaningful blocker when it is not schedulable;

? show the affected resource, window, movement or downstream commitment;

? preserve manual override and approval lineage;

? compare candidate options when scenario and recovery capability is activated.

5.1 Capability provided by the base scaffolding

Capability	Role in the ABL build
Scheduling entity model	Represents shipment movements, route steps, dependencies, priorities and target dates
Route and resource eligibility	Matches each movement with valid jetties, routes, tug-barge pairs, CTS assets and operating steps
Capacity-window placement	Places movements into free operating windows and prevents invalid overlap
Current-state loading	Considers active work, occupied resources, queues and confirmed actuals before placing new work
Dynamic constraint evaluation	Rechecks feasibility when tide, bridge, asset, berth or cargo readiness changes
Scenario engine	Supports delay, outage, window-change, reassignment and recovery assumptions after adoption gates are met
Candidate comparison	Evaluates alternate paths and recovery options against feasibility and operational impact
Exception and blocker output	Identifies the operating blocker and the affected downstream commitments

5.2 ABL-specific configuration

The build will configure the scaffolding around:

- ? Berau OGV demand, laycan, cargo quantity, coal grade and loading sequence;
- ? ABL jetties, BLC capability, tug-barge fleet, CTS assets and route timings;
- ? tide windows, bridge windows, river movement rules and safe crossing logic;
- ? cargo readiness, source eligibility and quality-release status;
- ? active trip state, loaded barges, queues and confirmed operational events;
- ? exception categories, override rules, approvals and publishability checks;
- ? reporting, dashboard, audit, export and integration requirements.

6. Product Scope: Phase-Gated Build Model

The application build is structured as a phase-gated maturity path. The first client-owned product is the governed scheduling MVP. Later blocks expand the product only when the operating model, user adoption and data readiness support the next level of automation.

Stage	Sprint direction	Client-owned product	Deferred by design
Stage 1 - Governed scheduling MVP	Sprint 0 to Sprint 5	Demand intake, cargo sequence, master data, operating windows, assignments, conflicts, approvals, publish and export	Live telemetry dependency, autonomous recovery, full commercial optimization
Stage 1A - Operational adoption gate	After Release 1 pilot usage	Evidence of planner usage, data gaps, overrides, false positives and approval behavior	Expansion before operating fit is validated
Stage 2 - Exception and replanning support	Sprint 6 onward, subject to gate	Exception taxonomy, delay propagation, manual recovery options and impact review	Autonomous recovery publication
Stage 3 - Scenario and impact simulation	Subject to scenario readiness	Baseline-versus-scenario comparison, projected downstream impact and utilization review	Scenario use without agreed data and decision rules
Stage 4 - Telemetry-assisted execution	Subject to integration readiness	Position/status evidence, event candidate workflow and confirmed event updates	Treating raw telemetry as automatic truth
Stage 5 - Governed flow-management maturity	Later maturity state	Guided recovery, management review surfaces and projection indicators	Final commercial settlement unless separately scoped

7. Release 1 - Governed Scheduling MVP

Release 1 proves the governed scheduling spine and operating-window logic. This release falls between **Sprint 0 and Sprint 5** of the ten-sprint build plan.

7.1 Sprint coverage

Sprint	Build focus	Key outcome

Sprint	Build focus	Key outcome
Sprint 0	Application foundation: web application, API service, database, background processing, object storage and routing layer	Runnable product spine
Sprint 1	Master data and governance: organizations, roles, audit, locations, cargo, jetties, routes, assets and compatibility	Safe multi-party operating model
Sprint 2	Demand and capacity intake: OGV demand, cargo layer, asset windows, jetty windows, tide windows and bridge windows	Structured inputs for planning
Sprint 3	Scheduling entity and route model: trips, assignments, schedule events and route-step dependencies	Movement chain represented as schedulable work
Sprint 4	Capacity placement and feasibility: candidate paths, occupied/free windows, conflict checks and movement candidates	Feasible draft plan with explicit blockers
Sprint 5	Plan governance: versions, overrides, approval requests, approval decisions, published snapshots and exports	Approved execution contract

7.2 Client deliverables in Release 1

Deliverable type	What the client receives
System screens	Login, role-based navigation, master data, OGV demand, cargo layer sequence, tide/bridge windows, asset availability, tug-barge assignment, jetty/BLC loading, CTS operations, conflict review, approval, published plan, audit and export screens
Scheduling capability	Structured demand intake, route/resource eligibility, candidate movement generation, capacity-window placement, feasibility checks, conflict output and manual override capture
Governance capability	Plan versioning, dual approval, publishability check, immutable published snapshot, audit trail and governed export
Product owned at this stage	A working scheduling application that replaces spreadsheet-only planning for the core ABL-Berau flow and creates an approved operating schedule

7.3 MVP success criteria

Success measure	Validation direction
Publishable-plan cycle time	Time from demand intake to publishable plan is reduced or made more predictable
Schedule rework	Repeated spreadsheet revisions and manual reconciliation are reduced
Conflict clarity	Hard blockers carry a reason code, affected movement and next review action
Approval traceability	Published plans carry approval, version and audit lineage
Planner adoption	Named pilot users can complete representative planning workflows with acceptable override usage
Data readiness	Demand, master data and operating-window inputs meet minimum completeness expectations

Specific numeric targets will be baselined with ABL and Berau during pilot usage.

8. Operational Adoption Gate

Before expanding beyond the governed scheduling MVP, ABL, Berau and Vector will review whether the tool matches real planning behavior.

The adoption gate will review:

? planner usage across representative scheduling workflows;

? demand completeness and master-data stability;

? conflict false positives and missed operational constraints;

? manual override frequency and reason-code quality;

? approval latency and escalation behavior;

? data gaps that prevent reliable scheduling;

? operating rules that need refinement before scenario or telemetry expansion.

The output of this gate is a clear decision on whether to move into exception support, scenario simulation, telemetry-assisted execution or additional workflow hardening.

9. Exception Taxonomy And Human Operating Reality

The application will preserve planner discretion during early adoption. Planners will still be able to handle negotiated exceptions, operational judgment and manual escalation. The difference is that these decisions will be captured with reason codes, approval state and plan lineage.

During pilot usage, repeated manual decisions will be reviewed to determine whether they should become formal scheduling rules, exception categories or operator guidance.

9.1 Initial exception taxonomy

Exception family	Initial examples	Release 1 treatment
Cargo/source readiness	Cargo not ready, quality hold, source switch, cargo layer mismatch	Conflict reason or readiness note
Jetty/BLC operations	Loading delay, queue, reduced loading rate, equipment stoppage	Capacity-window conflict or manual override
Navigation constraint	Tide missed, bridge window missed, draft restriction change, crossing risk	Operating-window blocker
Fleet/asset availability	Tug unavailable, barge unavailable, incompatibility, breakdown, charter lead-time pressure	Resource availability or eligibility blocker
CTS/transshipment	CTS queue, floating crane downtime, discharge rate reduction	Downstream capacity blocker

Exception family	Initial examples	Release 1 treatment
OGV/laycan	OGV ETA change, hatch sequence change, laycan pressure, loading completion risk	Demand priority and completion-risk note
Weather/marine	Rain, wind, visibility, wave restriction, river condition	Exception note or window blocker
Survey/clearance	Draft survey delay, documentation hold, port clearance interruption	Closure, readiness or handover blocker
Human/governance	Approval delay, manual override, escalation, disputed decision	Workflow and audit event
Data/integration	Missing feed, stale signal, duplicate event, untrusted telemetry	Data quality warning

9.2 Human-operational model

Release 1 will support human operation through:

- ? manual override with mandatory reason capture;
- ? planner notes and exception comments where operational judgment is required;
- ? approval workflow before publication;
- ? shadow-mode validation where rules need observation before enforcement;
- ? review of recurring overrides as candidates for future rule formalization.

10. Later Maturity Blocks

Later maturity blocks are part of the product direction, but they are not treated as dependencies for the first scheduling MVP.

Release block	Capability	Activation condition
Release 2 - Exception and manual recovery support	Delay propagation, exception queue, recovery option review and manual replanning path	Adoption gate confirms planning fit and exception taxonomy
Release 3 - Scenario and impact simulation	What-if assumptions, baseline-versus-scenario comparison, projected trip/event impact and utilization review	Data quality and scenario decision rules are agreed
Release 4 - Telemetry-assisted execution	GPS/AIS-style evidence, latest asset state, geofence/ETA variance, event candidates and confirmed event workflow	Integration owners, data trust and operating confirmation rules are ready
Release 5 - Governed flow-management maturity	Guided recovery, management review surfaces, publishability checks and projection-only operating impact indicators	Recovery governance and decision authority are stable

10.1 Economic and operating impact indicators

The first release prioritizes feasible and publishable scheduling. Commercial optimization is not treated as final settlement logic. Where data is available, later maturity blocks can expose projection-only indicators such as:

- ? laycan risk;
- ? waiting time;
- ? fleet utilization;
- ? queue impact;
- ? charter lead-time pressure;
- ? demurrage exposure proxy.

Final demurrage, invoicing, despatch and settlement remain outside the scheduling MVP unless separately scoped.

11. Working Flow Details

11.1 Clean planning flow

```
Import OGV demand
-> Review cargo grade and layer sequence
-> Enter operating windows
-> Generate movement candidates
-> Generate draft plan
-> Review conflicts
-> Apply reason-coded override if required
-> Submit approval
-> Complete Berau / ABL approvals
-> Run publishability check
-> Manually publish
-> Generate governed export
```

11.2 Capacity placement logic

At planning time, the platform will:

1. load current demand and master data;
2. plot current work and already occupied capacity;
3. identify eligible resources for each movement;
4. generate candidate route and resource paths;
5. search for continuous feasible operating windows;
6. mark selected windows as occupied;
7. record conflicts where no feasible placement exists;
8. preserve reason codes, override notes and audit lineage.

11.3 Disruption flow in later maturity blocks

```
Open active exception
-> Review operating cause
-> Generate recovery options
-> Compare impact
-> Create scenario from selected option
-> Simulate downstream effect
-> Promote feasible scenario
-> Repair remaining blockers if required
-> Submit approval
-> Manually publish recovery plan
```

11.4 State separation

The application will keep five states separate:

State	Meaning
Planned	The approved schedule expectation
Observed	Field signal or external evidence that may indicate status
Confirmed	Operationally trusted event that updates actual execution status
Projected	Future impact estimated by the scheduling or scenario engine
Recommended	Candidate action for review, not an automatic publication

12. Application Modules

12.1 Core MVP modules

Module	Working responsibility
Organizations and access control	Multi-party users, roles, permissions, data scope and approval authority
Master data	Locations, coal grades, mines, stockpiles, jetties, tug/barge/CTS assets, routes, loading rates and compatibility
Planning intake	OGV voyages, cargo requirements, cargo layer sequence, availability windows, tide windows, bridge windows and import jobs
Scheduling	Plan, plan version, trip, assignment, schedule event, conflict, override, approval, publish and export
Audit	Request and domain audit trail for governed actions
Operator cockpit	Dense operational screens for demand, cargo sequence, assignments, jetty/BLC, CTS, tide/bridge, conflicts, approvals and published plan

12.2 Later maturity modules

Module	Working responsibility
Scenario	Delay, outage, window-change and reassignment assumptions with projected impact
Telemetry	Source registry, asset mapping, position evidence, geofence zones, latest state, ETA projection and tracking alerts
Operations events	Integration feeds, device endpoints, health snapshots, event candidates, confirmed events and actualization
Recovery guidance	Recovery input snapshots, ranked options, root-cause review, scenario creation and publishability assessment
Management review	Operating impact, utilization, exception pattern and projection-only commercial indicators

13. Proposed Implementation Architecture

The implementation stack is the infrastructure behind the operator experience. Operators interact with workflows, screens, approvals and schedules; the stack provides the reliability, testability and deployment structure required to run those workflows.

Layer	Technology	Role in the application
Operator web application	React, Vite, TypeScript	Builds the browser-based screens for planners, approvers and operations users
Web application testing	Vitest, Testing Library, jsdom	Tests user interface behavior, screen logic and component interaction before release
API and business services	Django, Django REST Framework	Provides business APIs, workflow rules, access control and application services
Background jobs	Celery, Celery Beat	Runs asynchronous jobs such as imports, exports, scheduled checks and periodic processing
Application server	Gunicorn	Serves the backend application in a production deployment pattern
Service testing	pytest, pytest-django	Tests business services, data rules, API behavior and scheduling logic
Database	PostgreSQL with PostGIS	Stores operational data and supports location-aware/geospatial data where required
Cache / broker	Redis	Supports fast shared state and message brokering for background jobs
Object storage	MinIO / S3-compatible storage	Stores imported files, exports, generated documents and object artifacts
Edge routing	Nginx	Routes web and API traffic, supports deployment hardening and reverse-proxy needs
Runtime approach	Containerized deployment model	Packages services into repeatable deployment units for local, UAT and production environments

13.1 Integration technology positioning

Release 1 does not depend on live industrial integration. Where required by later scope, the integration layer can use:

? HTTP/API callbacks for business-system events;

? MQTT for lightweight device or event messaging;

? NMEA or AIS-derived feeds for marine position data;

? OPC-UA or PLC/operator-console integration for industrial equipment status;

? batch file exchange where operational readiness is lower than API readiness.

14. Data Sources And Integration Readiness

The first release can operate with manual entry, controlled upload and validated seed/replay data. Live integration should be activated only when data ownership, timing, reliability and operating response are agreed.

External data source	Assumed content	Release 1 handling	Later integration direction
Existing schedule files	Current OGV plan and operating schedule	Import or controlled upload	Governed file ingestion or direct integration
OGV schedule	ETA, ETB, laycan, quantity, priority and vessel status	Manual/imported demand	API feed from Berau or shipping system
Cargo/source data	Mine, CPP, stockpile, grade, product, available quantity, quality release and jetty eligibility	Master data and demand setup	Cargo, quality or ERP integration
Cargo layer sequence	Hatch/layer order, grade requirement and substitution rule	Cargo layer model	Structured demand feed
Jetty/BLC data	Loading rate, queue, working hours, equipment status and readiness	Manual windows/status	Terminal, PLC or operator-console feed
ABL fleet data	Tug, barge, CTS capacity, compatibility, status, location and availability	Master data and availability windows	GPS/AIS, Spinergie-style feed or fleet API
Tide data	Tide tables, height, safe movement windows and water-level observations	Manual tide windows	Sensor/API feed
Bridge data	Opening schedule, crossing rule and queue	Manual bridge windows	Bridge operator console or device feed
Weather/marine data	Rain, wind, wave, visibility and closure alerts	Exception input or planning note	Weather API or local station
GPS/AIS data	Position, speed, heading, timestamp, signal quality and external asset identity	Not required for Release 1	Tracker, AIS receiver, vendor API or NMEA-derived source
Operational events	Loading start/end, discharge start/end, stoppage, breakdown and crossing confirmation	Operator-entered confirmation or pilot replay	MQTT, HTTP callback, PLC, weighbridge, tablet or edge gateway
Survey and documents	Draft survey, sampling, certificates, SOF and final quantity/quality documents	Deferred milestone visibility	Document management or survey integration
Commercial parameters	Laycan risk, waiting-time proxy and exposure proxy	Deferred from MVP	Separate commercial sign-off

15. Build Deliverables

15.1 Release 1 deliverables

Deliverable	Description
Vector scheduling base configuration	ABL-specific setup of proprietary scheduling and dynamic-constraint scaffolding
Application foundation	Containerized web application, API service, database, Redis, object storage and routing layer
Master and capacity model	Assets, routes, eligibility, calendars, operating windows and compatibility
Demand and cargo intake	OGV demand, laycan, cargo quantity, coal grade and cargo sequence
Scheduling configuration	Shipment movement entity, candidate paths, capacity placement, moving-window checks and conflict output
Operator screens	Role-aware screens for demand, master data, windows, assignments, conflicts, approvals, publish and export
Governance layer	Access control, dual approval, publishability, immutable snapshots and governed exports
Evidence and test pack	Service tests, web application tests, proof commands, UAT scripts and runbook

15.2 Later maturity deliverables

Deliverable	Description
Scenario configuration	What-if assumptions, projected impacts, utilization views and scenario comparison
Exception and recovery support	Exception queue, impact review, recovery options and proof pack
Telemetry-assisted execution	GPS/AIS-shaped evidence, latest state, geofence/ETA variance and data trust review
Operations event layer	Event candidates, confirmations, rejection, actual event updates and device/feed health
Management review surfaces	Exception patterns, operating impact, projection-only indicators and maturity reporting

16. Assumptions, Readiness Gates And Boundaries

16.1 Assumptions

1. Berau provides OGV demand, laycan, cargo quantity, coal grade and layer sequence in a consistent format.
2. ABL and Berau agree master-data naming for jetties, routes, assets, grades, locations and operating zones.
3. Tide and bridge windows are initially entered manually or uploaded before live integration is activated.
4. Resource capacity includes both time availability and operational eligibility.

5. Active work in progress is plotted before new schedules or recovery options are generated.
6. Field evidence does not directly change schedule authority without confirmation.
7. Published plans are immutable; replanning creates successor versions.
8. Approval requires agreed ABL and Berau authority before publication.
9. Manual override requires reason capture and audit.
10. External chartered assets require readiness and lead-time assumptions.

16.2 Readiness gates

Gate	Required validation
Release 1 readiness	Core demand, master data, operating windows and approval users are available
Adoption gate	Pilot users can create, review, approve and publish schedules with acceptable manual intervention
Scenario gate	Delay, outage and reassignment assumptions are agreed and data quality is sufficient
Telemetry gate	Integration owner, feed reliability, identity mapping and confirmation rules are agreed
Recovery gate	Exception taxonomy, approval path and recovery decision authority are stable

16.3 Build boundaries

1. Release 1 is a governed scheduling MVP, not a live control tower.
2. Live GPS/AIS is evidence, not automatic truth.
3. Raw pings do not update the execution plan directly.
4. Confirmed field events update actual execution state only after the confirmation rule is satisfied.
5. Recovery options are advisory until simulated, reviewed, approved and checked for publish readiness.
6. Commercial projection is not final settlement logic.
7. Customer-facing visibility comes after internal governance is stable.
8. Planned, observed, confirmed, projected and recommended states remain separate.

17. Glossary

Term	Definition
ABL	The operating organization responsible for transshipment logistics coordination in this proposal context.
AIS	Automatic Identification System, a marine tracking signal used to identify vessel location, heading, speed and related navigation information.
BLC	Barge Loading Conveyor or barge loading capability used at jetty/loading points for coal movement into barges.
Berau	Berau Coal, the shipment-demand context for the proposed ABL operating blueprint.

Term	Definition
Bridge window	A permitted time range for a tug-barge movement to pass a bridge based on opening rules, navigation safety or operating coordination.
Candidate movement	A possible scheduled movement generated by the planning logic before it is accepted into a draft or published plan.
Cargo layer sequence	The required sequence in which coal grades or cargo layers are loaded into an OGV hold or hatch.
Constraint	A limiting factor that affects whether a movement can be planned, such as asset availability, tide timing, bridge opening, jetty capacity or cargo readiness.
CTS	Coal Transshipment Station or transshipment asset used for floating transfer/discharge operations between barges and ocean-going vessels.
Demurrage exposure proxy	A planning indicator that estimates potential commercial exposure due to waiting or delay. It is not final settlement logic.
Dynamic constraint	A constraint whose value or effect changes during operations, such as tide level, bridge availability, equipment status or asset position.
Feasibility check	A schedule validation step that confirms whether a movement can be placed within eligible resources and valid operating windows.
Geofence	A virtual geographic boundary used to detect whether an asset has entered, exited or remained within an operating zone.
Governed scheduling spine	The controlled planning workflow from demand intake through schedule generation, conflict review, approval, publish and export.
Jetty	The loading point where cargo is transferred to barges before movement toward CTS or OGV operations.
Laycan	The contractual or operating window within which an OGV is expected to be ready for loading.
Manual override	A planner action that allows a schedule decision to proceed despite a blocker or rule exception, with reason and audit capture.
MQTT	Message Queuing Telemetry Transport, a lightweight publish/subscribe protocol commonly used for IoT, device and event-message integration.
NMEA	National Marine Electronics Association data format family used by marine navigation equipment, often carrying GPS, AIS and vessel instrument data.
OGV	Ocean Going Vessel, the receiving vessel for the final coal-loading operation.
OPC-UA	Open Platform Communications Unified Architecture, an industrial communication standard used to exchange structured machine, equipment and process data.
Operating window	A time range within which an activity is permitted or feasible, such as tide movement, bridge crossing, jetty loading or CTS discharge.
Operator console	A local system, screen or control interface used by operational staff to enter, confirm or view equipment and process status.
PLC	Programmable Logic Controller, an industrial controller used to monitor and control equipment such as conveyors, loading systems and related plant operations.
Planned state	The approved schedule expectation before field execution changes are confirmed.
Publishability check	A final validation that a plan is ready to be released as an approved operating schedule.

Term	Definition
Published plan	An approved, immutable schedule snapshot that becomes the operating reference until a successor version is approved.
Reason code	A standardized explanation attached to a conflict, exception or manual override.
Recovery option	A candidate corrective action proposed for review after an operational disruption.
Scenario	A what-if schedule view that tests assumptions before changing the published plan.
Shadow-mode validation	A pilot approach where rules are observed and measured before being strictly enforced in live operations.
Tide window	A time range during which water level and navigation conditions allow a safe or permitted movement.
Tug-barge pair	A tug and barge combination considered together for movement eligibility, availability and compatibility.
UAT	User Acceptance Testing, the client-side validation stage for workflows, screens and outputs before production use.

18. Reference Inputs

This extension is based on:

? the Operational Blueprinting & Simulation Prototype Proposal;

? the Scheduling Simulation BRD;

? ABL and Berau Coal operating context;

? observed transshipment planning and fleet-coordination challenges;

? cargo, route, asset, tide, bridge, CTS and OGV scheduling logic;

? the product-development direction for a governed ABL operational scheduling platform.

End of application build extension